

Vons: A Compact, Host-Controlled Decision Component for Agent Workflows

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Abstract

Agents built around frontier language models repeatedly make small, bounded decisions: which of a few candidate actions to take, whether a precondition holds, or whether to ask before acting. We ask whether a compact local model can serve such a decision inside a host-controlled workflow, and what breaks under the constraints we tested. Vons is a decision component consisting of a bounded decision contract, a shared frozen encoder (a four-layer BERT) with two decision heads—a one-pass *Direct* head and a four-step DDIM *Diffusion* head—and a host-side adapter in which the model only proposes and the host alone validates, authorizes and executes. All measurements come from one hash-verified evidence packet, and the empirical findings are mostly negative. On a 200-row synthetic test split every answerable row shares one label, so the Direct head’s answerable accuracy of 1.0 equals that of a constant predictor; the Diffusion head reached 0.011–0.060. A post-hoc temperature fit on a disjoint calibration split stopped at its search boundary. On the original Mind2Web test splits, 37,512 exact- k selection attempts produced no accepted selection: 84.7% exceeded the 512-token input budget and the remainder fell below an uncalibrated confidence gate. The positive findings are narrow engineering results: a shared bundle with both heads occupies 46,001,618 bytes (43.9 MiB), under a 64 MiB design target; its CPU graph partition matched the original graphs exactly in 80 of 80 comparisons; and on CPU, padding Direct inputs from 64 to 512 tokens added 204.4 ms at the median, whereas the two heads differed by 1.3 ms at matched shape. We claim no external generalization, no superiority of either head, no calibrated deployment and no production readiness; we report the design, the evidence discipline and the failure modes so that they can be built upon.

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1 Introduction

A frontier-model agent spends much of its budget on decisions whose answer space is small and known in advance: pick one of up to 32 candidate elements, confirm that a precondition holds, or decide to ask for clarification. Serving such a step with a compact local model is attractive for cost, latency and privacy, and it is a natural complement to routing and cascading schemes that send only hard queries to a large model [1], [2]. It also raises three separable questions.

1. **Capability.** Does the compact model choose as a reference would on the decisions that actually occur?
2. **Cost.** How large are the model assets, and how long does a request take on the target runtime? For Vons that runtime is a browser running ONNX Runtime Web [3] with a WASM or WebGPU execution provider.

3. **Authority.** Can the model’s output ever cause an action that the host did not authorize?

Vons answers the third question by construction: the model only proposes, and the host derives every consequential action (Section 3). The first two questions are empirical. This paper reports what one frozen evidence packet supports on them and gives the negative results the same weight as the positive ones.

Contributions.

1. A bounded decision contract and a host-side adapter (KASI) in which confirmation is derived from host-owned risk metadata and exact-call consent, never from the model (Section 3).
2. A shared-encoder model with two decision heads, packaged as a single model-only bundle under a 64 MiB design target, with exact CPU graph-partition parity (Section 4 and Section 6.4).
3. A CPU shape decomposition showing that input padding, not the choice of head, dominates request time at the tested shapes (Section 6.5).
4. Informative negative results: a synthetic split that cannot distinguish learning from a constant predictor, a calibration fit that stopped at its boundary, and a Mind2Web diagnostic in which no selection was accepted because most inputs exceeded the token budget (Section 6.1 to Section 6.3).
5. An evidence discipline in which every reported number is bound to a hash-verified source field, with measured, historical, derived, design-target, planned and unavailable quantities kept apart (Section 5.1 and Appendix B).

What we do not claim. We do not claim that Vons generalizes to external tasks, that either head is better, that the calibration holds beyond the synthetic split, that the browser latencies reflect the current software, or that the system is ready for production use.

2 Related Work

Tool-using and web agents. ReAct interleaves reasoning traces with actions in language models [4], and Toolformer teaches a model to emit API calls inline [5]. Mind2Web provides tasks on real websites with crowdsourced action sequences for generalist web agents [6]; SeeAct grounds a multimodal model’s plans into element-level actions on the same benchmark [7], and WebArena supplies self-hosted environments for end-to-end evaluation [8]. Our external diagnostic uses the original Mind2Web test release. Vons addresses a narrower sub-problem than these systems: a single bounded choice inside a host-controlled loop, where the surrounding planner is out of scope.

Authority and risk in tool use. ToolEmu emulates tool execution to surface the risks that agents incur when they act with real consequences [9]. Vons takes the complementary, structural position that the model should never hold execution authority: the host owns the tool registry, the risk metadata and the consent binding, and the model’s output is advisory (Section 3). We present this as a design property of the frozen source, not as a measured safety outcome.

Compact encoders and distillation. Knowledge distillation transfers a large model’s behaviour into a smaller one [10]; DistilBERT and TinyBERT apply it to BERT-style encoders [11], [12], and pre-training remains important for compact models [13]. Vons uses one such compact pre-trained encoder [13], [14] without task-specific distillation. No local model was admitted as a distillation teacher in this work (Section 6.7).

Diffusion decoding. Denoising diffusion models [15] and the non-Markovian DDIM sampler that permits few-step generation [16] have been applied to discrete outputs through continuous relaxations [17]. The Vons Diffusion head applies four DDIM steps to candidate scores; its configuration is a first attempt with known design limitations (Section 4), and its results should not be read as a verdict on diffusion decoding.

Calibration and abstention. Temperature scaling is a single-parameter post-hoc calibration method [18]; expected calibration error (ECE) and the Brier score are standard summaries [19], [20]. Selective classification trades coverage for risk through a reject option [21], [22], and the maximum softmax probability is a common abstention signal [23]; language models’ own confidence has also been studied [24]. Vons reports coverage and selective risk alongside accuracy for this reason.

Vendor workflow evaluations. TypeSafe publishes a four-category workflow evaluation whose reference labels average GPT-6 Astra and Claude Fable 5.1 at high thinking [25], [26]. Scores there measure agreement with model-derived references rather than independently labelled truth. No comparison with that evaluation has been run for Vons, and no vendor figure is reproduced here.

3 The Decision Contract and the Host Boundary

Vons is a decision component, not an agent. A host application (for example, an agent runtime that uses a frontier model for planning) reaches a point where it must choose among a small, explicit set of options. It sends Vons a structured request. Vons returns a probability over the options, an answerability estimate, and either a choice or an abstention. The host then decides what, if anything, to do.

General decision contract. A request carries a structured state and 1–8 typed questions, each with 2–32 options, under a 512-token input budget. The response carries a choice with a probability vector and an answerability estimate, or an abstention. Score-type questions are not supported by the current candidate-selection ONNX runtime; an expected-ordinal score output is planned for an evaluation adapter and is not part of the implemented response. The Python validator estimates the token budget from characters, whereas the TypeScript SDK uses a tokenizer budget, so the two implementations do not yet form one harmonized public contract.

Host boundary (KASI adapter). For tool-using workflows, a separate host-side adapter called KASI maps the general contract onto five public actions: *call*, *clarify*, *confirm*, *refuse* and *respond*. KASI is kept separate from the general contract. Table 1 summarizes who decides what. The default confidence threshold is 0.55, and the Python adapter delegates the derivation to a single contract function. These are properties of source code and documentation; no adapter audit log from a live workflow exists, so boundary behaviour in deployment is UNAVAILABLE.

Table 1. Division of authority in the KASI host adapter. Implemented in the frozen source; not measured on live workflows.

Decision	Made by	Rule in the frozen source
Propose a call, clarification, refusal or response	Model	Proposal only; never executed by the model
Whether a tool may be used	Host	Unregistered tools are refused
Risk of a tool	Host registry	Missing or unknown risk fails closed
Whether to act at all	Host	Confidence below the threshold yields <i>clarify</i>
Whether confirmation is needed	Host	Required unless both the host-registered and the supplied advisory risk rank below <i>high</i>
Consent	User, via host	Bound to the complete call, including arguments
Execution	Host	The host is the only execution boundary

4 Model and Implementation

Encoder and heads. Both heads share the frozen encoder `google/bert_uncased_L-4_H-256_A-4` [13], [14] at a pinned revision (Appendix B), with a 512-token maximum and 32 option slots. The *Direct* head scores all candidates in one pass. The *Diffusion* head denoises candidate scores with four DDIM steps [16]. The historical Diffusion head has three known design limitations: it conditions on the pooled representation

only, it uses a weak 32-step terminal noise schedule, and it applies a softmax over denoised one-hot scores. Its results therefore do not establish that diffusion decoding is inferior in general.

Software. The implementation consists of a Python package (contract, data generation, training, evaluation, calibration, bundle manifest, KASI adapter and Mind2Web tooling), a TypeScript SDK (contract validation, KASI host policy and a browser runtime on ONNX Runtime Web; the historical browser runs recorded version 1.30.0), and exported ONNX graphs.

Shared bundle. The *shared-v1* bundle partitions the two existing full graphs into one encoder graph and two head graphs, with no training or parameter change (Section 6.4). Its recorded release status is “research candidate; parity and notices required before release”.

5 Evaluation Protocol

5.1 Evidence basis and labelling

All measurements come from one frozen evidence packet (schema `vons.publication-evidence/v2`, frozen 2026-09-24). Before drafting, we verified the packet archive and manifest digests, all 131 listed files by SHA-256 and byte count, and all 258 claim rows supplied with the packet. We also recomputed the six synthetic-pilot metric summaries from the saved predictions and test labels; the largest absolute difference from the recorded values was 2.2×10^{-16} . The packet records `git_revision = null` because the workspace had no Git repository at the 2026-09-24 freeze; provenance therefore rests on content hashes (Appendix B). A repository has since been designated (Section 10), but no later commit is bound to any measurement reported here.

We did *not* re-run training, model inference, browser tests or the Mind2Web evaluation. The restricted Mind2Web test data and the model weights are deliberately absent from the packet; their manifests, hashes and aggregate reports are present. Our Mind2Web statements therefore rest on verification of the aggregate reports, not on independent re-evaluation.

Throughout, quantities carry one of the following labels. **MEASURED**: read from a named field of a hash-verified report. **MEASURED (HISTORICAL)**: measured on an earlier configuration that is not the current software. **DERIVED**: arithmetic on measured fields, recomputed for this paper. **DESIGN TARGET**: an intended property, never an outcome. **PLANNED**: intended future work, not implemented or not run. **UNAVAILABLE**: not measured, or recorded as null in the source. **AUTHOR DECISION**: a licensing or status choice made after the evidence was frozen. Values in tables are rounded to the precision shown; the full-precision values are in the named packet fields.

5.2 Synthetic pilot data

The synthetic test split has 200 rows. A separate calibration split, filtered from the same generated source, also has 200 rows. Both splits have the same composition: 182 answerable rows, *all labelled call_tool*, 18 unanswerable rows, and four options per row. A constant `call_tool` predictor that never abstains would therefore reach answerable accuracy 1.0 on either split. Synthetic accuracy is not evidence of learned decision quality; we treat it only as a check that the pipeline runs end to end.

5.3 Metrics

Metrics follow the packet’s `metrics/v2` definitions.

- **Answerable accuracy**: correct covered predictions divided by all answerable rows; missing, error and abstained predictions count as misses.
- **Coverage**: covered predictions divided by all rows.
- **Selective risk**: errors divided by covered predictions; a covered prediction on an unanswerable row is an error.

- **ECE**: ten equal-width bins over covered predictions, computed on the recorded confidence. The recorded confidence is the maximum choice probability multiplied by the sigmoid of the answerability output, as computed in the frozen prediction loop; it differs from the choice probabilities used by Brier and NLL.
- **Brier** and **NLL** are means over a different population from ECE: the answerable rows whose prediction has no error and carries a valid probability vector. Abstained predictions are included when a valid vector exists. Per row, Brier is the multiclass score summed over the row’s options, $\sum_o (p_o - \mathbf{1}[o = \text{label}])^2$, so its range is 0–2; NLL is $-\log \max(p(\text{label}), 10^{-12})$. An empty population yields null (frozen `vons/evaluation.py`, metric definitions and `_nll/_brier`).

5.4 Mind2Web evaluation-only protocol

Data. We used the original Mind2Web test release [6] at a pinned dataset revision and repository revision (Appendix B). The licence is recorded as CC-BY-4.0 with research-only, no-training and evaluation-only restrictions. The train split was not acquired, so no train/test overlap audit exists. In total, 1,341 tasks and 9,378 action rows were normalized.

Retrieval and prediction. Candidates are ranked by lexical BM25 [27] ($k_1 = 1.2$, $b = 0.75$) over the released candidate pool, using the task goal and the teacher-forced official history strictly before the current action. The shared-v1 Direct model ran exact- k action-target selection for $k \in \{5, 10, 20, 32\}$, with a strict 512-token budget, no truncation, no calibration, and the historical uncalibrated gates of 0.55 confidence and 0.5 answerability.

Replay and intervals. A replay recomputes outcome counts from saved predictions. Task-macro intervals use 1,000 task-level bootstrap draws with seed 7. We report candidate recall and conditional selection accuracy separately. Browser task success and end-to-end DOM retrieval are not evaluated.

5.5 Timing protocols

CPU shape decomposition. This ran on CPU with ONNX Runtime 1.30.0 on macOS 26.6.2 (arm64), using the historical full graphs rather than shared-v1. The design used four synthetic cases, 20 repeats and 3 session starts, with 5 warm-up requests excluded per session, giving 240 samples per cell. The same live texts and masks were used across paired cells. Four cells were measured: (A) Direct at 4 slots $\times$ 64 tokens, (B) Direct at 32×64 , (C) Direct at 32×512 and (D) Diffusion at 32×512 .

Browser (historical). The historical Direct full graph was timed with 4 candidates in 4 padded slots at 512 tokens, with ten warm-up requests excluded per page/session start. The execution provider was *requested*, not observed: the observed provider is null and the requested provider is not proven. Fresh page sessions are not independent browser-process launches, and the page was not cross-origin isolated.

5.6 Local-model screen

Three local models (Qwen, Gemma and Muse Glimmer; Table 8) were each given the same 200 synthetic test rows under role metadata “writer”, “judge” and “analyst”. A later read-only audit bound all 600 responses to prompt and response hashes and made no new model calls. The three roles received identical prompt sets.

6 Results

6.1 Synthetic pilot

Table 2 reports the two heads separately; we do not rank them.

Direct. At every seed the Direct head covered all 200 rows and answered every answerable row correctly. Its selective risk of 0.09 is exactly the 18 covered unanswerable rows. Because of the single-label split, this shows only that the pipeline runs end to end.

Table 2. Synthetic pilot, 200-row test split, `metrics/v2` (MEASURED). The two heads are reported separately and are not compared: the split is single-label (Section 5.2), so answerable accuracy is uninformative about learning.

Head	Seed	Ans. acc.	Macro-F1	Coverage	Sel. risk	NLL	Brier	ECE
Direct	7	1.0000	1.0000	1.000	0.0900	0.1053	0.0139	0.2163
Direct	17	1.0000	1.0000	1.000	0.0900	0.0871	0.0097	0.1154
Direct	27	1.0000	1.0000	1.000	0.0900	0.1013	0.0129	0.2881
Diffusion (4 DDIM steps)	7	0.0275	0.0134	0.155	0.8387	1.8031	0.9318	0.4483
Diffusion (4 DDIM steps)	17	0.0604	0.0285	0.290	0.8103	1.8147	0.9370	0.4602
Diffusion (4 DDIM steps)	27	0.0110	0.0054	0.110	0.9091	1.8165	0.9368	0.5088

Diffusion. The Diffusion head covered 22–58 rows per seed and answered 2–11 answerable rows correctly. Its answerable accuracy of 0.0110–0.0604 is far below what a constant predictor attains on this split. This is a negative result for this configuration, not for diffusion decoding in general.

Head timing. The pilot reports also record decision-head latency with precomputed encoder embeddings. That quantity is neither request latency nor browser latency, and we do not report it as either.

6.2 Calibration (Direct, seed 7)

In the frozen source, calibration first fits a temperature and then fits an abstention threshold on temperature-scaled confidence for a design-target selective risk of 0.05, both on the separate calibration split (Table 3).

Table 3. Calibration fit and calibrated test result for the Direct head, seed 7. Values rounded to four decimals.

Quantity	Value	Label
Temperature-fit rows / threshold-fit rows	182 / 200	MEASURED
Fitted temperature (search range 0.05–20)	0.05 (at lower bound)	MEASURED
Abstention threshold	0.7558	MEASURED
Target selective risk	0.05	DESIGN TARGET
Calibration-split risk at threshold	0.0	MEASURED
NLL before / after (calibration split)	0.1064 / 0.0	MEASURED
Test coverage / selective risk / answerable accuracy	0.91 / 0.0 / 1.0	MEASURED
Test abstentions / covered unanswerable rows	18 / 0	MEASURED
Test ECE	0.2233	MEASURED

With the temperature at its lower bound the choice probabilities saturate: test Brier is 2.29×10^{-51} , test NLL is 0.0, and $1 - \overline{\max p}$ over covered rows is 0.0. Every covered test prediction is correct, so the calibrated ECE equals $1 - \overline{\text{confidence}}$ exactly, and that value is set by the answerability factor, not by the temperature. The 0.91 coverage corresponds exactly to abstaining on the 18 unanswerable rows.

This is an empirical result on one synthetic split for one seed of one head. It is not a risk guarantee and not calibration on any external task. Diffusion calibration is UNAVAILABLE, immutable checkpoint identity is absent from the prediction configurations, and calibrated browser parity is open.

6.3 Mind2Web: candidate recall and selection

Figure 1 and Table 4 show every split/ k cell.

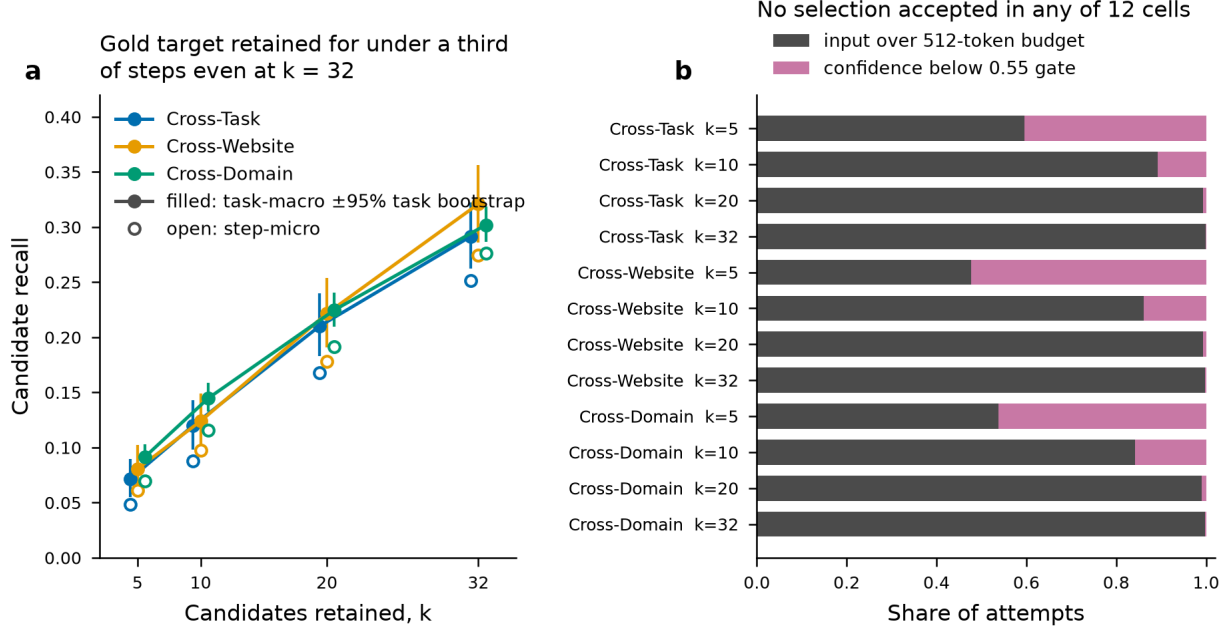


Figure 1. Mind2Web diagnostic on the original test splits (strict 512-token budget; uncalibrated 0.55/0.5 gates). (a) Candidate recall of the lexical BM25 retriever over the released pool; filled markers are task-macro recall with 95% task-bootstrap intervals, open markers are step-micro recall, and points are offset horizontally for legibility. (b) Outcome of every Direct selection attempt; accepted selections are zero in every cell.

Table 4. All twelve Mind2Web split/ k cells (MEASURED). “Overflow” means the input exceeded the 512-token budget; “abstain” means confidence fell below the 0.55 gate. Conditional selection accuracy is 0.0 in every cell. Rows per split (with a positive candidate in the pool): Cross-Task 2,094 (1,975); Cross-Website 1,373 (1,314); Cross-Domain 5,911 (5,590).

Split	k	Step-micro recall	Task-macro recall	95% CI (task-macro)	Recalled	Overflow	Abstain	Accepted
Cross-Task	5	0.0486	0.0713	[0.0549, 0.0894]	96	1,246	848	0
Cross-Task	10	0.0876	0.1199	[0.0983, 0.1429]	173	1,867	227	0
Cross-Task	20	0.1676	0.2100	[0.1833, 0.2397]	331	2,081	13	0
Cross-Task	32	0.2516	0.2916	[0.2623, 0.3223]	497	2,092	2	0
Cross-Website	5	0.0609	0.0806	[0.0615, 0.1025]	80	655	718	0
Cross-Website	10	0.0974	0.1240	[0.0992, 0.1489]	128	1,182	191	0
Cross-Website	20	0.1781	0.2216	[0.1908, 0.2537]	234	1,364	9	0
Cross-Website	32	0.2747	0.3210	[0.2860, 0.3561]	361	1,370	3	0
Cross-Domain	5	0.0698	0.0916	[0.0819, 0.1027]	390	3,180	2,731	0
Cross-Domain	10	0.1159	0.1447	[0.1328, 0.1587]	648	4,980	931	0
Cross-Domain	20	0.1916	0.2248	[0.2099, 0.2405]	1,071	5,852	59	0
Cross-Domain	32	0.2762	0.3021	[0.2869, 0.3188]	1,544	5,900	11	0

Totals. Across the twelve cells there were 37,512 selection attempts. Of these, 31,769 (84.7%) were input overflows and 5,743 were confidence abstentions; no record was missing and no selection was accepted.

Consequences. Conditional selection accuracy is 0.0 in every cell under the frozen metric, which keeps every recalled failure in the denominator, and acceptance coverage is 0.0. Complete-case conditional accuracy is undefined, because no selection was accepted. Each k has 499 rows without a positive candidate, counted separately.

Recall. Candidate recall rises with k . At $k = 32$, step-micro recall is 0.2516–0.2762 and task-macro recall is 0.2916–0.3210 across splits. This is a property of the lexical retriever and the released pool, not of the Vons model, and because history is teacher-forced it is not end-to-end retrieval.

Result. Under this protocol the shared-v1 Direct model is not useful on Mind2Web. The failure is dominated by the 512-token input budget: most attempts never reach inference.

6.4 Model assets and CPU graph parity

Table 5 lists the shared-v1 model-only bundle.

Table 5. Shared-v1 model-only bundle (MEASURED runtime and notices excluded).

Asset	Role	Bytes
encoder.onnx	encoder graph	553,978
encoder.onnx.data	encoder weights	44,423,168
direct.onnx	Direct graph	22,843
direct.onnx.data	Direct weights	132,096
diffusion.onnx	Diffusion graph	82,287
diffusion.onnx.data	Diffusion weights	74,752
tokenizer/tokenizer.json	tokenizer	711,396
tokenizer/tokenizer_config.json	tokenizer	379
calibration.json	calibration	233
config.json	configuration	486
Total model assets		46,001,618
Model-asset limit (64 MiB)	DESIGN TARGET	67,108,864

The shared bundle totals 46,001,618 bytes (43.87 MiB), below the 67,108,864-byte model-asset design target. Runtime bytes are null because no runtime is included. The two original full bundles are 45,853,779 bytes (Direct) and 45,899,284 bytes (Diffusion). Each is below the target on its own, but shipped together they total 91,753,063 bytes, which is 24,644,199 bytes over it (DERIVED).

Parity. Graph-partition parity compared the shared-v1 graphs with the full graphs on CPU (CPUExecutionProvider, ONNX Runtime 1.30.0) over 20 cases and 80 comparisons at $\text{atol} = \text{rtol} = 10^{-5}$. All 80 passed, with a maximum absolute difference of 0.0 for scores and answerability. The cases covered 2/4/8/16/32 candidates, batch sizes 1 and 2, and Diffusion noise seeds 7/17/27. This is graph-partition equivalence only; it says nothing about the tokenizer, the browser, calibration or latency.

6.5 CPU shape decomposition

Figure 2 and Table 6 give the four matched CPU cells.

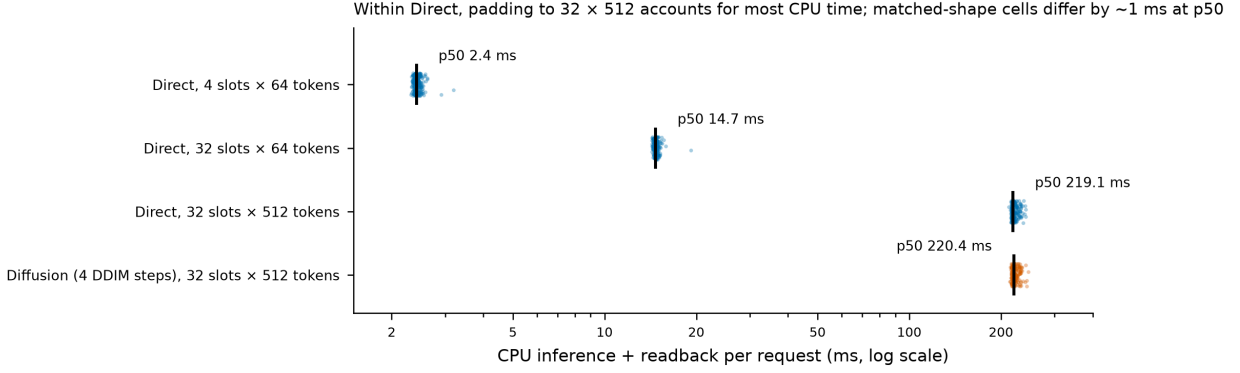


Figure 2. Per-request CPU inference plus readback time for four matched cells (240 samples each; four synthetic cases, three session starts, 20 repeats, five warm-ups excluded). Black ticks mark the median. Historical full graphs on ONNX Runtime 1.30.0 CPU; not a browser measurement.

Table 6. CPU shape cells, inference plus readback (MEASURED).

Cell	Configuration	Samples	p50 (ms)	p95 (ms)
A	Direct, 4 slots $\times$ 64 tokens	240	2.416	2.535
B	Direct, 32 slots $\times$ 64 tokens	240	14.705	15.154
C	Direct, 32 slots $\times$ 512 tokens	240	219.078	232.687
D	Diffusion (4 DDIM steps), 32 slots $\times$ 512 tokens	240	220.367	232.473

Within Direct, expanding 4 slots to 32 at 64 tokens adds 12.289 ms at p50 (B – A), and expanding 64 tokens to 512 at 32 slots adds 204.373 ms (C – B). The matched-shape difference D – C is 1.289 ms at p50 (all DERIVED). The latter is a system difference that still includes architectural differences; it is neither the cost of DDIM alone nor a method-only speed comparison. The decomposition covers four cases on one CPU, and matched-padding cells in the browser remain open.

6.6 Historical browser timings (Direct full graph)

Table 7 reports the only browser measurements in the packet. They are historical and use the Direct full graph, not the current SDK or the shared-v1 bundle.

Table 7. Historical browser measurements of the Direct full graph (MEASURED, HISTORICAL). Both columns use 4 candidates $\times$ 512 tokens, 500 timed samples, five page/session starts and ten warm-ups excluded per start. “Requested” means the execution provider was requested by the page, not observed.

Quantity	Requested WASM	Requested WebGPU
Successful / failed samples	500 / 0	500 / 0
Page/session starts	5	5
Inference + readback p50 / p95 (ms)	2207.0 / 2470.2	91.0 / 130.5
Total request p50 / p95 (ms)	2207.4 / 2471.0	91.4 / 132.0
Tokenize p50 (ms)	0.3	0.2
JS heap used after (bytes; JS-heap scope only)	11,811,939	57,884,604
Requested provider proven	false	false
Process-level cold start claimable	false	false

These are request configurations measured on one Mac. The executing provider and graph partition are unverified, so the difference between the columns cannot be attributed to WebGPU execution and no speed ratio is claimed. Process and GPU peak memory are UNAVAILABLE the JS-heap figures describe only that heap. The current SDK and shared-v1 bundle were not timed, the Diffusion head has no browser cells, and the one attempt at a browser numerical-parity matrix stalled without a raw report and is not a pass.

6.7 Local-model screen

Table 8. Historical local-model screen (MEASURED, HISTORICAL 200 synthetic rows each; no new model calls). ^a Whether the local model conversion carries a licence field; not a statement about upstream licence terms.

Model (local tag)	Role	Params, quant.	Rows	Strict JSON	Null on abstain	Known-label acc.	Licence field ^a
qwen3.8:27b-mlx	writer	27.8B, nvfp4	200	200	200	0.00	true
gemma4:31b	judge	31.3B, Q4_K_M	200	200	200	0.00	true
muse-glimmer:30b-mlx	analyst	32.3B, nvfp4	200	200	182	0.09	false

On the 182 answerable rows labelled `call_tool`, all three models chose `respond_directly`. The saved prompts neither identify a required tool nor state a rule that would make `call_tool` decidable, so these low accuracies do not rank the models. Muse Glimmer’s 0.09 comes from its 18 abstention flags; on those rows it kept a non-null *clarify* choice, so only 182 of its 200 rows satisfy the null-on-abstention contract. No output was used for training, no local model was admitted as a teacher, and the upstream revisions of all three local conversions are null.

7 Discussion

The input budget breaks first. The Mind2Web result points at the input contract before the model. With teacher-forced history and full candidate text, 84.7% of attempts exceeded 512 tokens before inference. On CPU the sequence budget is also the dominant cost: going from 64 to 512 tokens added 204.373 ms at the median, against 12.289 ms for going from 4 to 32 slots. Any retriever, truncation or chunking change is a new protocol and must not be tuned on these held-out results.

The synthetic benchmark cannot measure learning. Because every answerable row shares one label, the synthetic split cannot separate a learned policy from a constant one. The Diffusion head’s failure on the same split is informative only about this configuration, given its known design limitations.

Calibration is a pipeline, not a result. The calibration pipeline runs in the intended order on a disjoint split, but its temperature stopped at the search boundary and the calibrated ECE is set by the answerability factor. We therefore regard calibration as not established.

Direct versus Diffusion is not adjudicated. Quality is uninformative on this split, the matched-shape CPU difference is small but includes architecture, the browser has only Direct cells, and no equal-effort tuning record exists.

What holds, within scope. A shared-encoder bundle with both heads fits under a 64 MiB model-asset design target and partitions exactly on CPU. The Direct pipeline runs from synthetic data through calibration to evaluation. A historical Direct request configuration produced 500 successful timed samples per requested browser provider on one machine. The host boundary keeps consequential actions out of the model’s hands by construction, although its live behaviour is unmeasured.

8 Limitations

1. The only quality data for Vons heads are synthetic, and the answerable label is constant.
2. The Mind2Web diagnostic is evaluation-only and was not independently re-run. It uses a lexical retriever with teacher-forced history, and its result is dominated by the token budget.
3. Browser timings come from a historical graph under requested, unverified providers on one Mac, with no independent process launches and no process or GPU memory measurement.
4. CPU shape results cover four cases on one machine.
5. Calibration covers one seed of one head, and its temperature reached the search boundary.

6. The frozen packet carries no Git revision (`git_revision = null` at capture) and no immutable checkpoint identity for the pilot predictions; historical runtime code identity is limited wherever a raw report does not bind it, and no later repository commit can be bound to these measurements retroactively.
7. The public contract is not fully harmonized between Python and TypeScript.
8. No adapter audit log from live workflows exists, so KASI boundary behaviour in deployment is unmeasured.
9. The checks recorded by the implementer (241 Python tests passed with 50 subtests; lint exit 0; 10 SDK tests passed; browser, benchmark and parity builds exit 0) were not re-run for this paper, and unit or mocked tests do not establish browser numerical parity.

9 Open Gates and Future Work

Table 9 records the implementer’s gate status at the evidence freeze; it was not re-audited for this paper.

Table 9. Release-gate status recorded at the evidence freeze.

Gate	Recorded status
G0 bundle manifest and provenance	implemented and verified
G1 real WASM inference	implemented; parity harness ready; execution open
G2 WebGPU and memory	Direct controlled cells measured; matrix open
G3 shape and padding	CPU alternative implemented and measured; browser matched cells open
G4 Mind2Web	measured under evaluation-only authorization; public and training rights restricted
G5 attribution and training inclusion	preparation complete; inclusion open
G6 vendor workflow comparison	protocol frozen; comparison not run
G7 public release review	release hold
G8 distribution licence	proposal only at freeze; terms selected by the author afterwards (Section 10)
G9 browser extension channel	architecture proposal only

Open measurement work. Required next steps are: Diffusion browser cells; a completed browser numerical-parity matrix; process and GPU memory; timing of the current SDK and shared-v1; score support; DDIM step counts 2 and 8 with additional seeds; lifecycle and offline checks; matched-padding browser cells; complete-case accuracy and end-to-end task success on Mind2Web; a train-split overlap audit; an equal-effort Direct/Diffusion tuning record; Diffusion calibration; immutable checkpoint identity; live KASI audit logs; and the vendor workflow comparison.

Community participation (PLANNED). The author intends to open Vons to outside reading, trial, experience reports, reproduction and contribution. None of this exists yet as a measured activity: there are no external users, reviews, reproductions or hosted demos to report. Any future experience report would be anecdotal and would not count as research validation unless it was collected under a stated protocol with denominators.

10 Availability, Licensing and Disclosure

Article. This article and its original figures are licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>), which includes commercial reuse under its terms. The licence does not cover the Vons software, quoted source excerpts or third-party assets, raw benchmark data, or private verification records.

Software and models. At the evidence freeze, the Python package metadata declared Apache-2.0 and a proposed alternative licence had not been adopted; the TypeScript package had no licence field, and

the shared-v1 bundle carried no licence or notice payload. The release assessment recorded at the freeze is a hold on public release of the software and models (gate G7, Table 9), with every scientific gate open. Separately, and after the freeze (AUTHOR DECISION), the author selected source-only distribution of the Vons software under custom source-available terms: noncommercial use is free, and enterprise and commercial use require a separate agreement. These terms are not an OSI open-source licence and do not cancel grants already made under earlier terms; their licence text and notices are not part of the frozen evidence. Publishing source does not validate model deployment, and none of the results above changes with it.

Code and model locations (AUTHOR DECISION, access not verified). The author has designated <https://github.com/inlevel9-com/Vons>, <https://huggingface.co/INLEVEL9> and <https://huggingface.co/INLEVEL9/Vons> as the distribution locations. Commits, pushes and uploads are performed by the author personally, and links from the source repository README back to this paper and to the Technical Report remain withheld pending review. Their public-access and upload status could not be verified from the drafting environment; we make no claim about which files are available at any of them.

Data. Mind2Web data were used evaluation-only, were excluded from training and are not redistributed; only aggregate reports and hashes are held in the evidence packet. The release status of the synthetic pilot data is undecided. The frozen evidence packet is held by the author.

Models used and contributions. The encoder is `google/bert_uncased_L-4_H-256_A-4` [13]. Qwen, Gemma and Muse Glimmer were used only in the historical suitability screen, as comparison records rather than labels. For Muse Glimmer, the upstream model card and LICENSE state Apache-2.0, and a separate upstream usage policy lists prohibited uses and states that the model is not intended for individuals under 18 [28]. The local conversion has no licence field and no mapping to an upstream revision, so its exact upstream revision is not established, and we draw no legal conclusion. Implementation contributions are attributed only to recorded outputs: Claude Code (the `metrics/v2` and `calibration/v2` fixes, with a scoped check of 25 passed tests and 13 subtests); TRAE (a browser raw-sample summary tool accepted with 71 scoped tests, and a read-only licence inventory); a local Qwen model (one bounded static review of the Mind2Web predictor, whose three findings were dispositioned as not reproduced, rejected against an existing guard, and one accepted diagnostic improvement); and Codex (integration, the shared bundle and its parity, Mind2Web acquisition through replay, the browser verifier and orchestration).

Writing assistance. This manuscript was drafted with the assistance of Claude Opus 5.5 and revised for publication with the assistance of Claude Fable 5.1. The assistants verified hashes, recomputed aggregates from saved predictions, checked references and wrote text; they did not generate or alter any measured value. The author is responsible for the content.

A Superseded Metric Values

An earlier metrics version (v1) scored the same saved predictions differently: it ignored covered unanswerable rows in selective risk and divided the Brier score by the option count. Table 10 lists the superseded values next to the v2 values used in this paper (MEASURED, HISTORICAL).

Table 10. Superseded v1 and current v2 metric values on the same saved predictions.

Report	Selective risk v1 → v2	Brier v1 → v2	ECE v1 → v2
Direct, seed 7	0.0000 → 0.0900	0.0035 → 0.0139	0.3008 → 0.2163
Direct, seed 17	0.0000 → 0.0900	0.0024 → 0.0097	0.1991 → 0.1154
Direct, seed 27	0.0000 → 0.0900	0.0032 → 0.0129	0.2479 → 0.2881
Diffusion, seed 7	0.8214 → 0.8387	0.2330 → 0.9318	0.4327 → 0.4483
Diffusion, seed 17	0.8000 → 0.8103	0.2343 → 0.9370	0.4537 → 0.4602
Diffusion, seed 27	0.9091 → 0.9091	0.2342 → 0.9368	0.5088 → 0.5088

B Provenance and Traceability

Table 11 lists the identifiers that pin the evidence used in this paper. Table 12 maps each section, table and figure to the evidence-packet files it rests on. Row-level bindings (file, JSON pointer or source line, SHA-256 and value) are kept in the author’s audit ledger, which is available for review on request; the *Key* column is the ledger key.

Table 11. Provenance identifiers of the frozen evidence (verified before drafting).

Item	Identifier
Evidence archive publication-evidence-v2.zip, SHA-256	f2fb06c2275a60eec6c5b89938123da599b9979bb10d7e65a5667f6ca69152c4
Packet manifest packet.json, SHA-256	7cc5b19cd0e7691919d632d431ff1101d5a5b26a7c11c104c9fa83c132cd5318
Packet schema and freeze time	vons.publication-evidence/v2; 2026-09-24T08:41:25.776377+00:00
Encoder and revision	google/bert_uncased_L-4_H-256_A-4 at 387825ce42dbb39b87911cdf8e383ee3b25184f8
Synthetic test split, SHA-256	e5fdd950a35593b97d9799435a8739177c1668d2ad7d30687546add0bc4de1bc
Synthetic calibration split, SHA-256	a986ab870f9d54d3905db1f3c71ea4e004922a5bdf43fe822e4c503d7dd3689
Mind2Web dataset and dataset revision	osunlp/Mind2Web at 17ece8eb89862368edc0cc806acee6fca5163474
Mind2Web repository revision	33bd95caeee7bba22dd08ecc935845e15c5e5dc7
Mind2Web test archive, SHA-256	8f5fbc72afab942fe97cdf7fb397e179885d89b5c16862288e9a14bc6d41ca89
CPU runtime and platform	ONNX Runtime 1.30.0; macOS-26.6.2-arm64-arm-64bit

Table 12. Traceability map from manuscript locations to evidence-packet sources.

Key	Location	Evidence sources
M01	Abstract: synthetic pilot	reports/quality-summary-v2.json; data/generated/pilot-test.jsonl
M02	Abstract: calibration	reports/pilot-direct-test-calibrated-v2.json; data/generated/pilot-calibration.manifest.json
M03	Abstract: model assets and parity	artifacts/shared-v1/manifest.json; artifacts/provenance/browser-direct/bundle-manifest-v1.json; artifacts/provenance/browser-diffusion/bundle-manifest-v1.json; reports/shared-v1/parity.json
M04	Abstract, §6.6: browser scope	reports/browser-direct-wasm-trae-summary.json; reports/browser-direct-webgpu-trae-summary.json
M05	Abstract: Mind2Web totals	reports/mind2web/evaluation-replay-v1/summary.json
M06	Abstract: CPU shape decomposition	reports/onnx-shape-decomposition.json
M10	§3: general decision contract	vons/contract.py; sdk/typescript/src/onnx.ts; packet.json
M11	§3, Table 1: KASI host boundary	docs/KASI_INTEGRATION.md; docs/JEV_COMPARISON_PROTOCOL.md; vonskasi.py; docs/PUBLICATION_REVIEW.md; vonskcontract.py; packet.json
M12	§4: encoder, heads, shared bundle	reports/pilot-direct-test-calibrated-v2.json; docs/ORCHESTRATION_STATUS.md; packet.json; artifacts/shared-v1/manifest.json
M13	§2: vendor workflow evaluation status	evals.typesafe.ai; typesafe.ai blog; reports/quality-summary-v2.json; docs/JEV_COMPARISON_PROTOCOL.md; packet.json
M20	§5.1: verification and non-reruns	packet.json; reports/quality-summary-v2.json
M21	§5.2: synthetic data composition	reports/quality-summary-v2.json; data/generated/pilot-calibration.manifest.json; data/generated/pilot-test.jsonl; data/generated/pilot-calibration.jsonl
M22	§5.3: metric definitions	reports/quality-summary-v2.json; vonsevaluation.py; vonstraining.py; docs/PUBLICATION_REVIEW.md

Key	Location	Evidence sources
M23	§5.4: Mind2Web protocol	reports/mind2web/data-provenance-v1.json; reports/mind2web/evaluation-replay-v1/summary.json; reports/mind2web/prediction-run-v1.json; tools/predict_mind2web.py; packet.json
M24	§5.5: timing protocols	reports/onnx-shape-decomposition.json; reports/browser-direct-wasm-trae-summary.json; reports/browser-direct-webgpu-trae-summary.json; docs/ORCHESTRATION_STATUS.md
M25	§5.6: local-model screen protocol	reports/local-model-alignment-v1.json
T1, T2	Table 2: synthetic metrics	reports/quality-summary-v2.json
M30	§6.1: synthetic interpretation	reports/quality-summary-v2.json; reports/pilot-direct-test-calibrated-v2.json; data/generated/pilot-test.jsonl
T3	Table 3: calibration	reports/pilot-direct-test-calibrated-v2.json
M31, M47	§6.2: calibration interpretation and order	reports/pilot-direct-test-calibrated-v2.json; packet.json; docs/ORCHESTRATION_STATUS.md; vons/calibration.py
T4	Figure 1, Table 4: Mind2Web cells	reports/mind2web/evaluation-replay-v1/summary.json
M35	§6.3: Mind2Web totals and consequences	reports/mind2web/evaluation-replay-v1/summary.json; packet.json
T5	Table 5: shared-v1 bundle	artifacts/shared-v1/manifest.json
M32	§6.4: bundle and parity	artifacts/shared-v1/manifest.json; artifacts/provenance/browser-direct/bundle-manifest-v1.json; artifacts/provenance/browser-diffusion/bundle-manifest-v1.json; reports/shared-v1/parity.json
T6	Figure 2, Table 6: CPU shape cells	reports/onnx-shape-decomposition.json
M34	§6.5: shape differences	reports/onnx-shape-decomposition.json; packet.json
T7	Table 7: historical browser	reports/browser-direct-wasm-trae-summary.json; reports/browser-direct-webgpu-trae-summary.json
M33, M46	§6.6, §4: browser limits and runtime version	packet.json; internal review record (not public); docs/ORCHESTRATION_STATUS.md; browser summaries
T8	Table 8: local-model screen	reports/local-model-alignment-v1.json; artifacts/ollama-models.json
M36	§6.7: alignment audit	reports/local-model-alignment-v1.json
M40	§7: discussion	reports/mind2web/evaluation-replay-v1/summary.json; reports/onnx-shape-decomposition.json; docs/ORCHESTRATION_STATUS.md; packet.json
M41	§5.1, §8: provenance and recorded checks	packet.json; reports/orchestration-checkpoint-v2.json
M42	§10: models used and contributions	internal review record (not public); docs/ORCHESTRATION_STATUS.md; docs/MODEL_ATTRIBUTION.md; Muse Glimmer LICENSE and USAGE_POLICY (upstream); artifacts/ollama-models.json
M43	§9, Table 9: gates and open work	docs/COMPLETION_PLAN.md; packet.json
M44	Title page, §10: licensing and data	author decision after freeze; docs/PUBLICATION_REVIEW.md; pyproject.toml; packet.json; sdk/typescript/package.json; artifacts/shared-v1/manifest.json; reports/mind2web/data-provenance-v1.json
M45	Table 11: provenance identifiers	packet.json; reports/quality-summary-v2.json; data/generated/pilot-calibration.manifest.json; reports/pilot-direct-test-calibrated-v2.json; reports/mind2web/data-provenance-v1.json; reports/onnx-shape-decomposition.json
TA1	Table 10: superseded metric values	reports/pilot-direct-test.json; reports/pilot-direct-seed17-test.json; reports/pilot-direct-seed27-test.json; reports/pilot-diffusion-test.json; reports/pilot-diffusion-seed17-test.json; reports/pilot-diffusion-seed27-test.json; reports/quality-summary-v2.json

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